

ML-Test-01

Section A – Formula & Definitions (10 × 1 = 10 Marks)

01/08/25

1. Write the formula for Linear Regression.
2. Write the formula for Logistic Regression (Sigmoid function).
3. Write the formula for PCA (Principal Component Analysis) projection.
4. Write the formula for Normalization (Min-Max Scaling).
5. What is the output range of the sigmoid function?
6. In a table, which is horizontal — row or column?
7. Which algorithm is used for classification and works using distances? (a) Decision Tree (b) KNN (c) Linear Regression (d) SVM
8. What type of values does Linear Regression predict? (a) Classes (b) Binary values (c) Continuous values (d) Boolean
9. What is the shape of a confusion matrix for binary classification? (a) 1x2 (b) 2x2 (c) 3x3 (d) 4x1
10. Name any one distance metric used in KNN.

Section B – 2-Mark Questions (5 × 2 = 10 Marks)

11. What is the difference between Normalization and Standardization?
12. What is the purpose of PCA in Machine Learning?
13. Define True Positive and False Negative with an example.
14. Write the steps involved in training a Linear Regression model.
15. What does `train_test_split()` do in sklearn?
- 16.

Section C – 5-Mark Conceptual Questions (3 × 5 = 15 Marks)

Answer any 3 of the following:

17. Explain how Linear Regression works with a diagram.
18. Explain the working of KNN (K-Nearest Neighbors) algorithm step-by-step.
19. What is a Confusion Matrix? Draw a 2x2 confusion matrix and explain TP, TN, FP, FN.
20. Write a short note on Decision Tree Algorithm.
21. What is the difference between Linear and Logistic Regression?

Section D – Coding (1 × 15 = 15 Marks)

21. Build and evaluate a Linear Regression model using the following dataset:

Data:

Experience = [1, 2, 3, 4, 5]

Salary = [30000, 35000, 40000, 45000, 50000]

Tasks:

1. Convert the data into a Pandas DataFrame.
2. Train a LinearRegression model using sklearn.
3. Predict the salary for 6 years of experience.
4. Plot the regression line.
5. Print model coefficient and intercept.